

- ② $P(a)$ is True, $Q(a)$ is False. By defn. of existential quantifier, $\exists x P(x)$ is True. By defn. of OR, $\exists x P(x) \vee \exists x Q(x)$ is True.
- ③ $P(a)$ is False, $Q(a)$ is True. By defn. of existential quantifier, ~~$\exists x P(x)$~~ $\exists x Q(x)$ is True. By defn. of OR, $\exists x P(x) \vee \exists x Q(x)$ is True.

$$\boxed{\therefore \exists x (P(x) \vee Q(x)) \text{ is True} \Rightarrow \exists x P(x) \vee \exists x Q(x) \text{ is True}}$$

Let, $\exists x P(x) \vee \exists x Q(x)$ be True. By defn. of OR, the following 3 cases arise:
 ① $\exists x P(x)$ and $\exists x Q(x)$ ^{are} True (both). By defn. of existential quantifier, there exists at least one element a_1 in the domain such that $P(a_1)$ is True. By defn. of existential quantifier, there exists at least one element a_2 in the domain such that $Q(a_2)$ is True. By defn. of OR, $P(a_1) \vee Q(a_1)$ is True and also $P(a_2) \vee Q(a_2)$ is True. By defn. of existential quantifier, $\exists x (P(x) \vee Q(x))$ is True.

② $\exists x P(x)$ is True and $\exists x Q(x)$ is False. By defn. of existential quantifier, there exists at least one element a_1 in the domain such that $P(a_1)$ is True. By defn. of OR, $P(a_1) \vee Q(a_1)$ is True. By defn. of existential quantifier, $\exists x (P(x) \vee Q(x))$ is True.

③ $\exists x P(x)$ is False and $\exists x Q(x)$ is True. By defn. of existential quantifier, there exists at least one element a_2 in the domain such that $Q(a_2)$ is True. By defn. of OR, $P(a_2) \vee Q(a_2)$ is True. By defn. of existential quantifier, $\exists x (P(x) \vee Q(x))$ is True.

$$\boxed{\therefore \exists x P(x) \vee \exists x Q(x) \text{ is True} \Rightarrow \exists x (P(x) \vee Q(x)) \text{ is True}}$$

$\therefore$ They are logically equivalent $\square$

Exercises 48-51 establish rules for null quantification that we can use when a quantified variable does not appear in part of a statement.